

Monetary Policy Transmission in the United States: A Structural VAR Analysis

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1 Research Question

This essay investigates whether U.S. monetary policy, as conducted through the federal funds rate (FFR), has a significant causal effect on two key macroeconomic indicators of social welfare: real output growth and inflation. Concretely, the question is whether innovations in the FFR – interpreted as exogenous monetary policy shocks – generate economically and statistically meaningful responses in real GDP growth and inflation, and whether these responses match the predictions of standard monetary transmission theory.

2 Theoretical Framework

The empirical workhorse used to answer this question is the Vector Autoregression (VAR), the standard tool for characterizing the joint dynamics of macroeconomic time series without imposing strong a priori restrictions on the underlying structural relationships (Stock and Watson, 2001). Let $y_t = (\Delta \ln RGDP_t, \Delta \ln P_t, FFR_t)'$ denote the vector of the three variables of interest – quarter-on-quarter real GDP growth, quarter-on-quarter inflation (based on the GDP price index), and the level of the federal funds rate. The reduced-form VAR(p) representation is

$$y_t = c + B_1 y_{t-1} + B_2 y_{t-2} + \cdots + B_p y_{t-p} + u_t, \quad u_t \sim (0, \Sigma_u), \quad (1)$$

where u_t is a vector of reduced-form residuals with variance-covariance matrix Σ_u . Because the elements of u_t are in general contemporaneously correlated, they cannot be interpreted as structural, economically meaningful shocks on their own. To recover such shocks, the structural form is written as

$$A_0 y_t = c^* + A_1 y_{t-1} + \cdots + A_p y_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim (0, I), \quad (2)$$

with $u_t = A_0^{-1} \varepsilon_t$ and $\Sigma_u = A_0^{-1} (A_0^{-1})'$. Since A_0 has more free parameters than Σ_u , identifying restrictions are required. This exercise follows the recursive (*triangular*, Cholesky) identification scheme: A_0^{-1} is restricted to be lower triangular, with variables ordered as GDP growth $\rightarrow$ inflation $\rightarrow$ FFR. This ordering imposes the standard short-run restriction that monetary policy shocks affect output and inflation only with a one-period lag, while the FFR is allowed to respond contemporaneously to shocks in output and inflation – consistent with the idea that the central bank observes (or can react within the quarter to) current economic conditions when setting the policy rate, whereas private-sector quantities adjust to policy with a delay.

It is worth noting that this constant-parameter framework imposes a single, time-invariant transmission mechanism over the entire sample. As shown by Primiceri (2005), the relationship between monetary policy and the macroeconomy may in fact vary over time, both in the size of shocks and in the systematic response of policy to the economy. The present exercise abstracts from this possibility for tractability, and this is flagged as a limitation in the discussion below.

3 Data and Methodology

Quarterly data from 1954Q3–2026Q1 were obtained from the FRED database: real GDP (series GDPC1, the continuation of the now-discontinued GDPC96), the GDP chain-type price index (GDPCTPI), and the effective federal funds rate (FEDFUNDS, aggregated to quarterly frequency by within-quarter averaging). Real GDP and the price index were transformed into quarter-on-quarter log-differences (multiplied by 100); the FFR was kept in levels, as specified in the assignment.

The lag order p was selected over the grid $\{1, 2, 3\}$ using the Schwarz/Bayesian Information Criterion (BIC), which selected $p = 1$ (BIC values of -2.864 , -2.854 , and -2.752 for $p = 1, 2, 3$ respectively). As a robustness check (extra credit), a rolling-origin, one-step-ahead forecast cross-validation exercise was performed over the same lag grid. The resulting mean squared forecast errors (2.305, 2.340, and 2.399 for $p = 1, 2, 3$) also favor $p = 1$, so the information criterion and the out-of-sample cross-validation procedure agree. This agreement suggests that, for this particular dataset and lag grid, model-selection uncertainty is not a major source of concern for the structural inference that follows – a second, alternative-lag IRF analysis was therefore not necessary.

The estimated VAR(1) is stable (the moduli of all characteristic roots are below unity: 0.964, 0.800, 0.064). A Portmanteau test for residual autocorrelation, however, rejects the null of white-noise residuals ($\chi^2(63) = 140.9$, $p < 0.001$), indicating that some short-run dynamics are not fully captured at $p = 1$. Since the assignment restricts the lag search to $\{1, 2, 3\}$, this is reported as a limitation of the exercise rather than corrected by deviating from the prescribed specification.

4 Results: Impulse Response Functions

Figure 1 presents the full 3×3 grid of orthogonalized impulse responses (20-quarter horizon, with bootstrapped 95% confidence bands), where each row corresponds to a structural shock and each column to the variable’s response.

GDP growth shocks. A positive output shock raises GDP growth on impact, but the effect dissipates within two to three quarters, consistent with the low estimated persistence of the growth equation. The shock has a small, short-lived positive effect on inflation. Most notably, it triggers a sharp and persistent *increase* in the FFR, consistent with a central bank that systematically tightens policy in response to stronger-than-expected output growth (Taylor-rule-type behavior).

Inflation shocks. An inflation shock has, by construction, no contemporaneous effect on GDP growth, and only a modest, not strongly significant negative effect thereafter. The shock displays strong persistence in inflation itself. The FFR response is positive from impact onward, building gradually to a peak around two years out – again consistent with a policy rate that responds systematically to inflationary pressure.

Monetary policy (FFR) shocks. This is the row of primary interest for the research question. A contractionary monetary policy shock has the theoretically expected – but only weakly significant

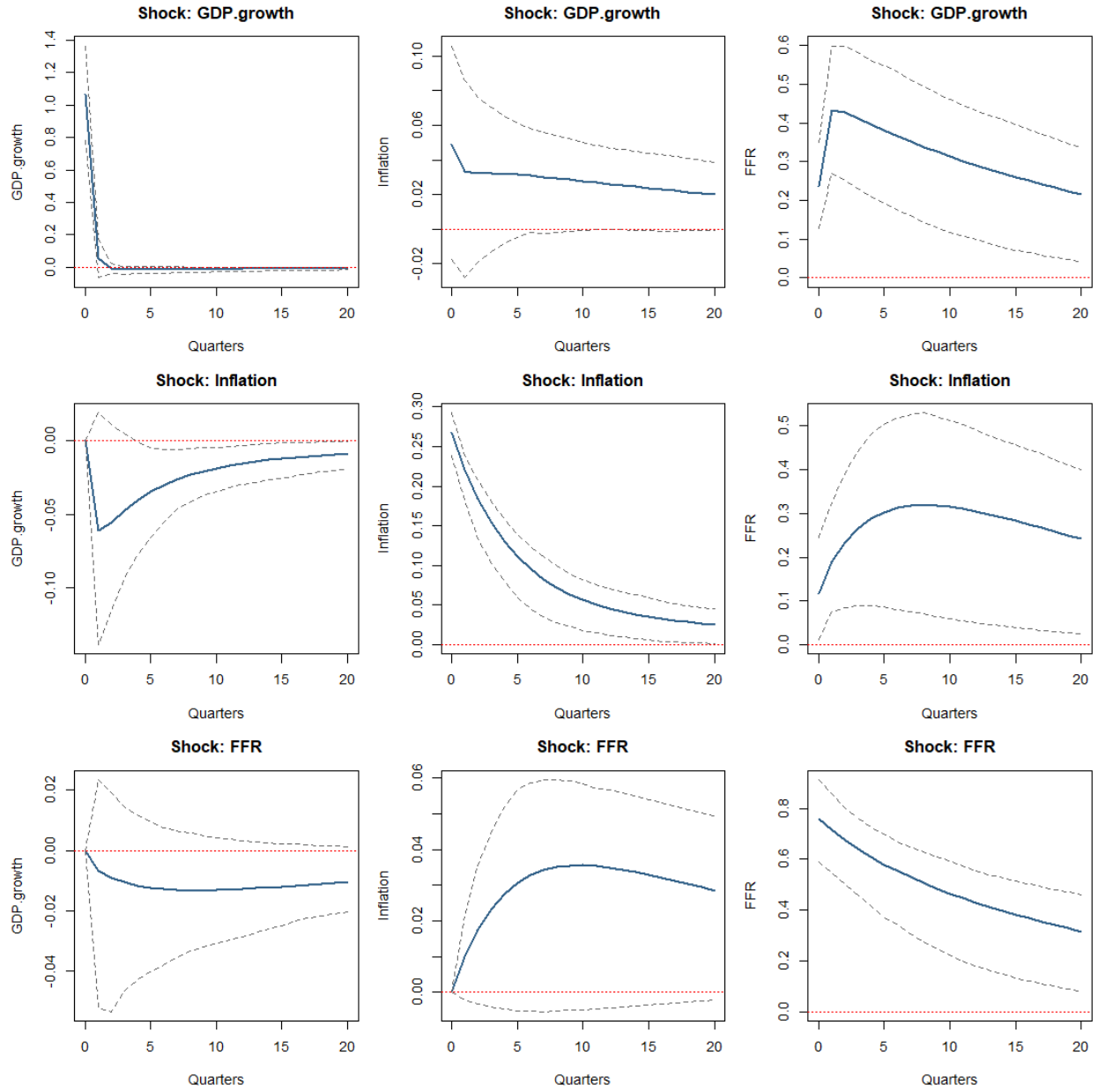


Figure 1: Orthogonalized impulse response functions, triangular (Cholesky) identification, ordering GDP growth → Inflation → FFR. Dashed lines are 95% bootstrap confidence bands.

– negative effect on output growth: the point estimate dips modestly and the confidence band includes zero for almost the entire horizon. More strikingly, the response of inflation to a monetary tightening is *positive* and statistically significant over an intermediate horizon (roughly quarters 3 to 17), peaking around two years after the shock. This is the well-known *price puzzle* documented extensively in the recursive SVAR literature on monetary policy. A standard explanation is that a purely backward-looking, recursively identified system omits any forward-looking indicator (e.g., inflation expectations or commodity prices) that the central bank uses to anticipate future inflation; when the Fed raises rates pre-emptively in response to information the econometrician does not observe, the model mechanically attributes the subsequent rise in inflation to the policy shock itself, rather than to the omitted information that triggered the policy response in the first place.

5 Conclusion

The evidence from this triangular SVAR is mixed with respect to the research question. On one hand, the FFR equation shows that monetary policy responds systematically and significantly to both output and inflation shocks, consistent with an active, rule-based central bank. On the other hand, the transmission *from* monetary policy shocks *to* the real economy is comparatively weak: the output effect is correctly signed but not strongly significant, and the inflation effect displays the well-documented price puzzle rather than the textbook disinflationary response. Information-criterion and cross-validation lag selection agree on $p = 1$, so these results do not appear to be an artifact of model-selection uncertainty within the considered lag grid. They are, however, conditional on the recursive identification scheme and on the constant-parameter assumption maintained throughout the sample, both of which are acknowledged limitations of this exercise relative to richer frameworks such as the time-varying SVAR of Primiceri (2005).

References

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